SDG 4.5.1 Gender Parity Dashboard

End User Guide

Table of Contents

1. Introduction
2. Dashboard Overview
3. Navigation Instructions
4. Using Interactive Elements
5. Understanding the Visualizations
6. Glossary of Terms
7. Data Sources
8. Limitations and Interpretation Guidelines

---

1. Introduction

Purpose

This dashboard analyzes gender parity in tertiary education and explores how education levels associate with quality of life outcomes across four European countries: Germany, Sweden, Serbia, and Spain. The analysis addresses UN Sustainable Development Goal 4.5.1, which measures equality of access to education.

Who Should Use This Dashboard

- Educational policymakers designing equity initiatives
- EU education authorities monitoring SDG 4 progress
- University administrators developing inclusive policies
- Researchers studying comparative education systems
- Students and families making educational decisions

What You'll Learn

- Whether gender parity exists in tertiary education attainment

- How education associates with income across different countries
- How education relates to life satisfaction and wellbeing
- Whether these relationships have changed from 2018-2023

---

2. Dashboard Overview

Dashboard Structure

The dashboard consists of **three pages**:

1. **Gender Inequality** - Examines gender parity among tertiary-educated individuals
2. **Income Gap** - Analyzes education-income relationships across countries
3. **Well-Being** - Explores education-life satisfaction associations

Data Coverage

- **Countries:** Germany, Sweden, Serbia, Spain
- **Time Period:** 2018, 2020, 2023
- **Sample Size:** ~32,000 respondents across all countries and years
- **Data Source:** European Social Survey (ESS) Rounds 9, 10, 11

---

3. Navigation Instructions

Accessing the Dashboard

Published Link: Introduction - SDG 4 Dashboard Valeri Kamenov 251041 - Power BI

Moving Between Pages

Option 1: Navigation Bar

- Click on any page name to jump directly:
    - "Introduction" → Page 1
    - "Gender Inequality" → Page 2
    - "Income Gap" → Page 3

- "Well-Being" → Page 4

First-Time User Recommended Path

1. Start with **Gender Inequality** page to understand parity status
2. Move to **Income Gap** page to see economic associations
3. Continue to **Well-Being** page for life satisfaction patterns

---

4. Using Interactive Elements

Country Slicer (All Pages)

Location: Bottom of Gender Inequality Page



Top of rest of pages



How to Use:

- **Select one country:** Click a single tile (e.g., "Germany")
- **Select all countries:** Click all four tiles or unselect all tiles

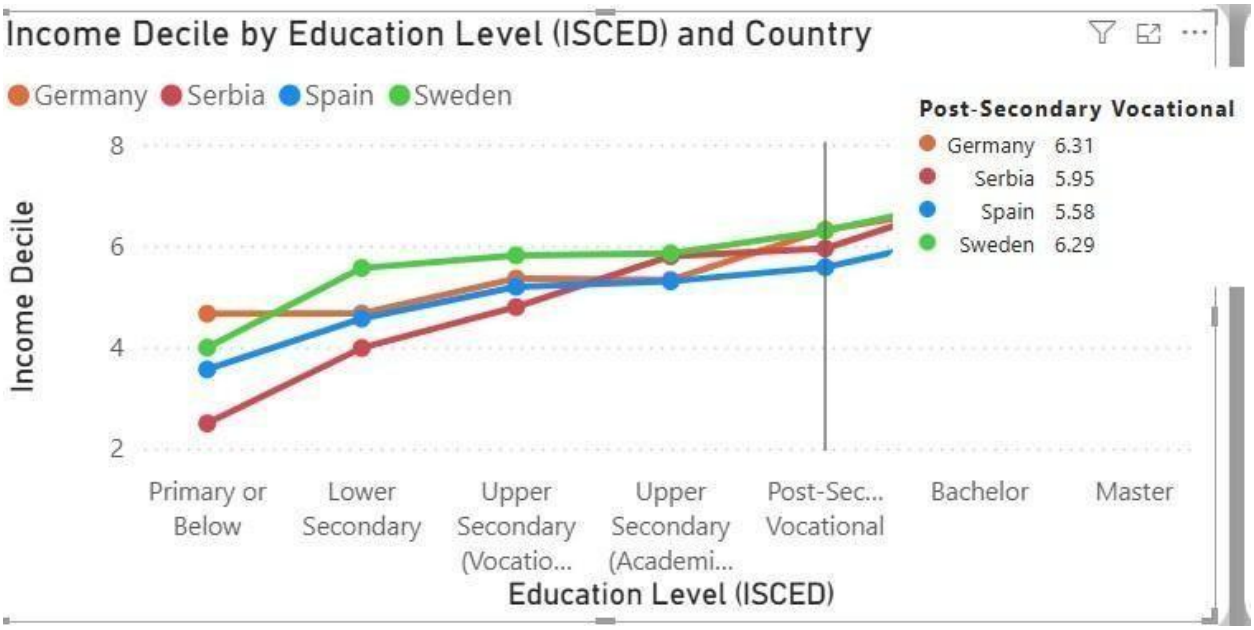
What Happens:

- All visualizations on the page update to show only selected countries
- KPI cards recalculate for filtered population
- Gap measures adjust to reflect selection

Use Cases:

- Compare Sweden vs. Serbia directly, for example
- Focus analysis on your home country, if it is present
- Present one country at a time during discussions

Tooltips (Hover for Details)



How to Use:

- Hover your mouse over any data point, bar, or line
- A tooltip appears showing exact values and additional context
- Move mouse away to hide tooltip

What You'll See:

- Exact numeric values (e.g., "Income Decile: 6.5")
- Category labels (e.g., "Female, 2020")
- Related metrics where applicable

---

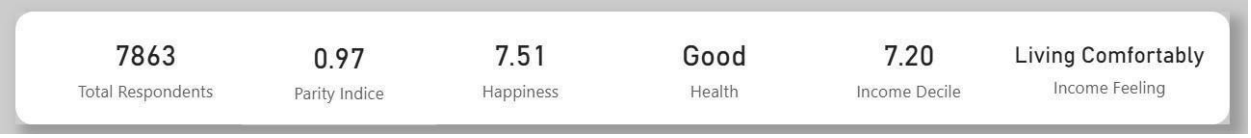
5. Understanding the Visualizations

Page 1: Gender Inequality

Purpose: Assess whether men and women have equal representation in tertiary education and comparable outcomes.

Filter Applied: This page shows only Bachelor and Master degree holders (tertiary education).

KPI Cards (Top Row)

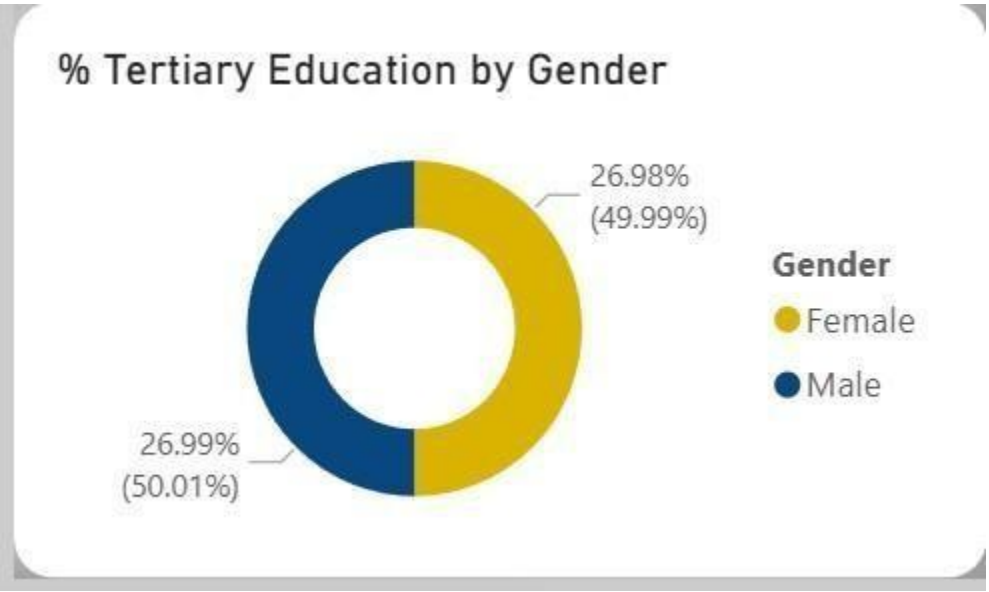


Five summary statistics showing:

- **Total Respondents:** Sample size for your selected country/countries
- **Average Happiness:** Mean score on 0-10 scale
- **Health Status:** Most common health category
- **Average Income Decile:** Mean income position (1=lowest, 10=highest)
- **Income Feeling:** Most common perception of household income

How to Read: These update when you change country selection. Use them to understand the characteristics of the filtered population.

Donut Chart: % Tertiary Education by Gender

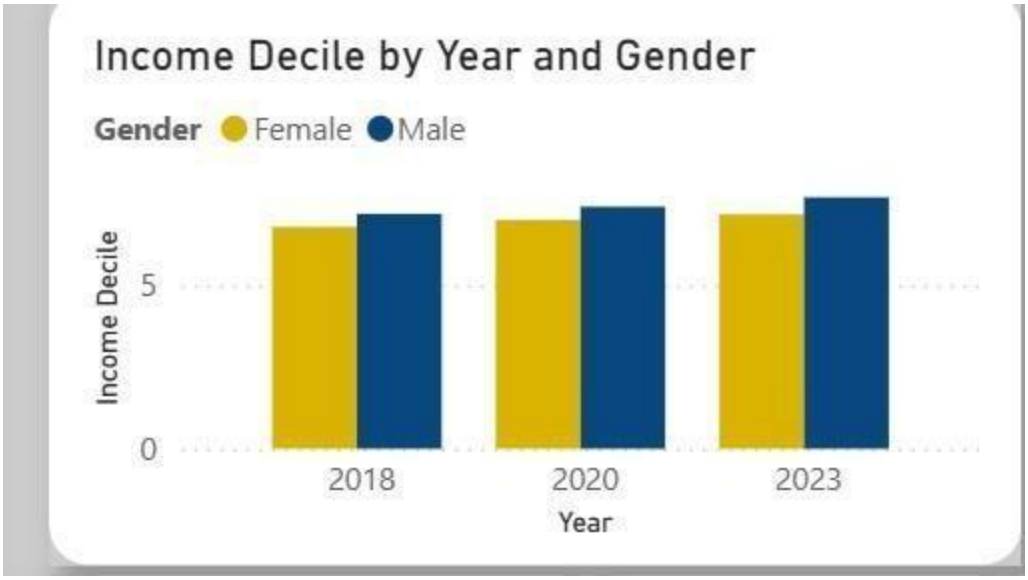


Shows: Proportion of men vs. women among tertiary-educated respondents

How to Read:

- Near 50/50 split indicates gender parity
- Larger female slice (51-52%) indicates slight female advantage
- Larger male slice would indicate male advantage

Clustered Column Charts (3 charts)



Show: Happiness, Income Decile, and Life Satisfaction compared between genders across three years (2018, 2020, 2023)

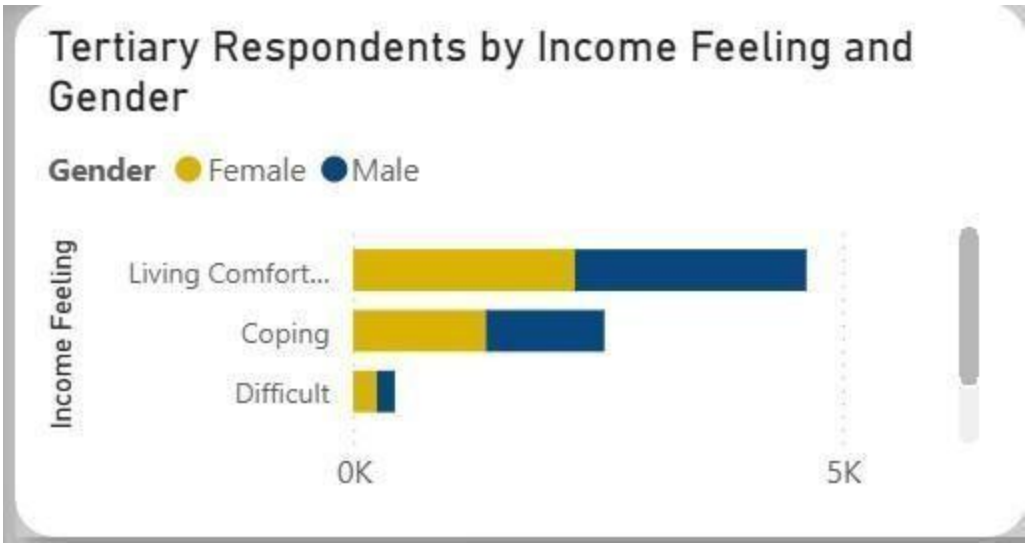
Colors:

- Yellow bars = Female
- Blue bars = Male

How to Read:

- Bars of similar height = similar outcomes between genders
- Taller male bars = men score higher
- Taller female bars = women score higher
- Look across years to see if gaps are changing over time

Stacked Bar Chart: Income Feeling by Gender



Shows: What proportion of men vs. women feel their household income is comfortable, coping, difficult, or very difficult

How to Read:

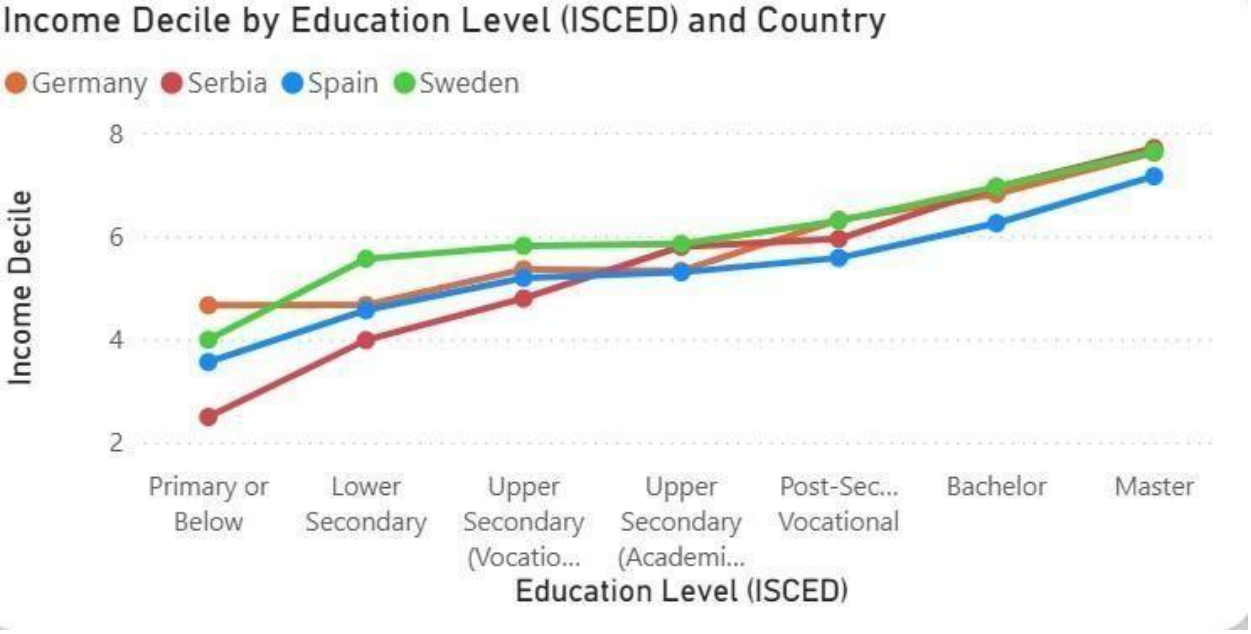
- Each bar represents total respondents of that gender's responses
- Segment sizes show proportions in each category
- Compare segment sizes between male and female bars
- Larger "Difficult" segments indicate more financial stress

Page 2: Income Gap

Purpose: Analyze how education level associates with income, and whether this relationship differs by country.

Filter Applied: None - all 7 education levels shown (Primary through Master).

Line Chart: Income Decile by Education Level and Country



Shows: Average income position (1-10 decile scale) for each education level in each country

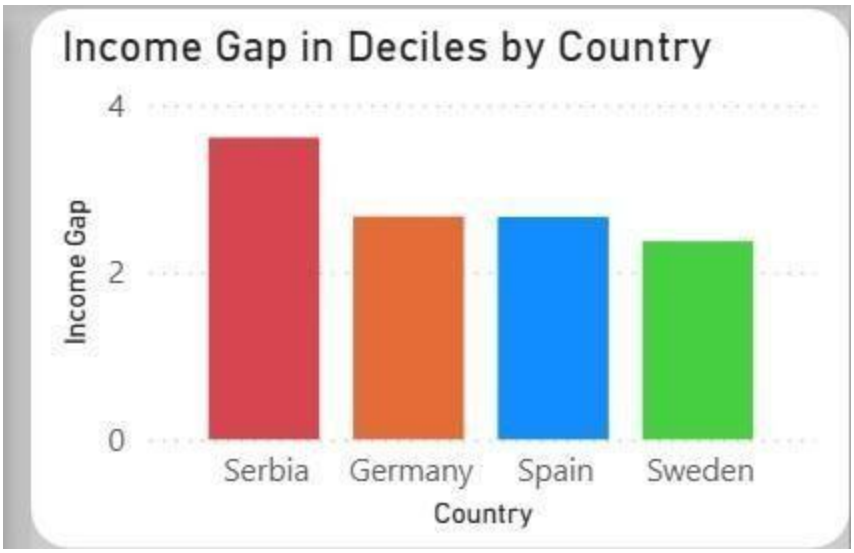
Lines:

- Four colored lines representing Germany, Sweden, Serbia, Spain
- Each line connects 7 points (one per education level)

How to Read:

- Steeper upward slope = stronger education-income association
- Flatter line = weaker association (income similar across education levels)
- Higher position = higher average income
- Compare line slopes to see which countries have strongest education-income relationships

Bar Chart: Income Gap by Country



Shows: Size of income gap between tertiary-educated and primary/lower secondary educated individuals

How to Read:

- Longer bars = larger gaps (education matters more for income)
- Bars measured in deciles (e.g., 3.5 = 3.5 income deciles difference)
- Countries ranked from largest to smallest gap

Gauge: Overall Income Gap

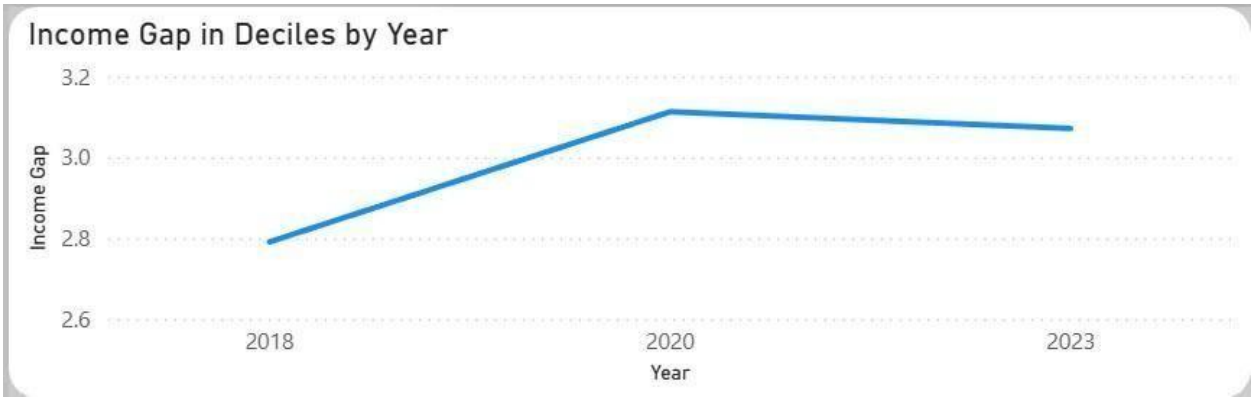


Shows: Average income gap across all four countries

How to Read:

- Needle points to gap size (measured in deciles)

Line Chart: Income Gap by Year

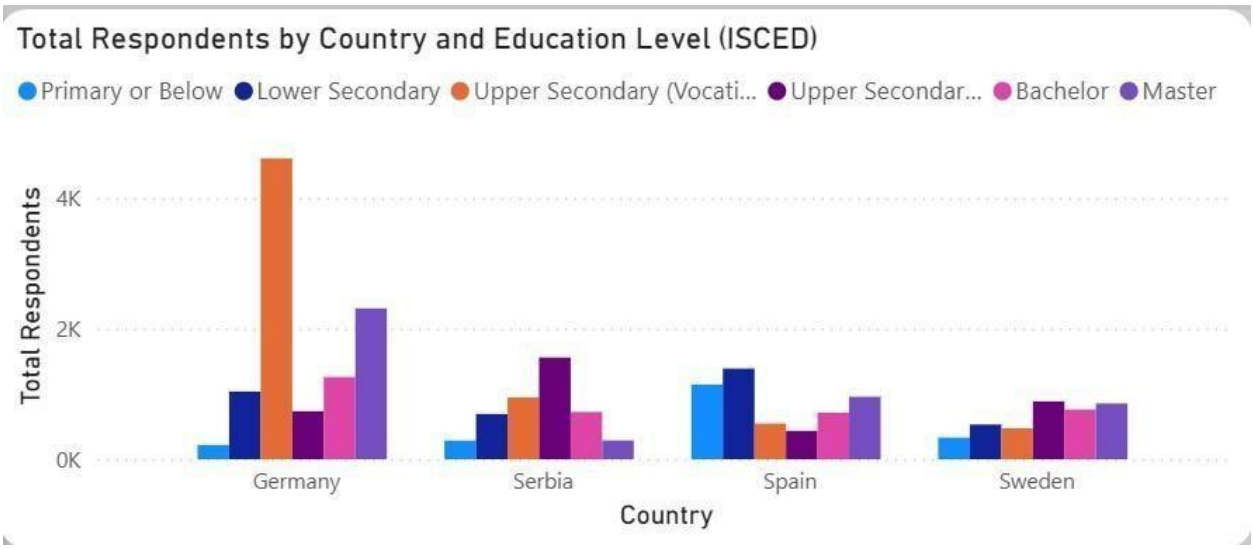


Shows: How income gaps changed from 2018 to 2023

How to Read:

- Upward slope = gap widening
- Downward slope = gap narrowing

Clustered Bar Chart: Education Distribution by Country



Shows: How many respondents of each country's population is at each education level

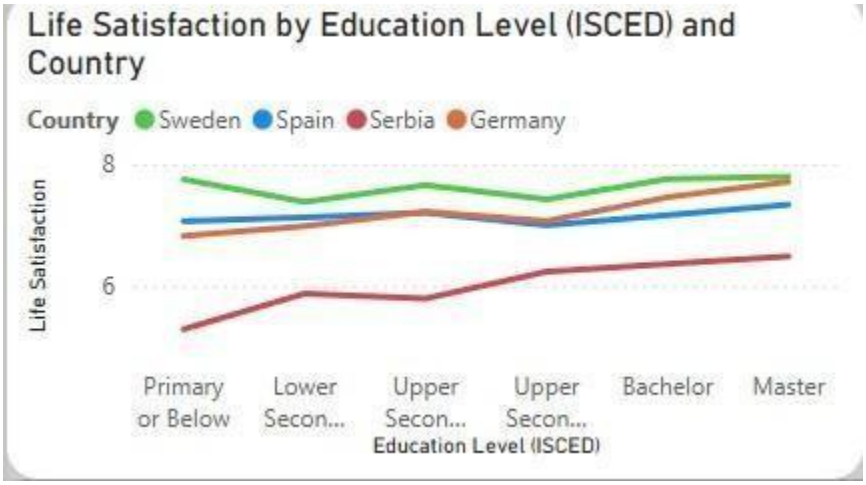
How to Read:

- Segment sizes show proportions at each education level
- Compare distributions to understand different national education systems

Page 3: Well-Being

Purpose: Examine how education associates with life satisfaction and other wellbeing indicators.

Line Chart: Life Satisfaction by Education Level and Country



Shows: Average life satisfaction (0-10 scale) for each education level in each country

How to Read:

- Similar to income line chart, but measuring life satisfaction instead
- Steeper slope = education more strongly associated with satisfaction
- Flatter line = satisfaction similar regardless of education level
- Higher position = higher average satisfaction

Bar Chart: Life Satisfaction Gap by Country

Shows: Difference in life satisfaction between tertiary-educated and primary/lower secondary educated

How to Read:

- Longer bars = larger gaps (education matters more for satisfaction)
- Measured in satisfaction points (0-10 scale)

Matrix Tables (3 tables)



Education Level (ISCED)	Health
Primary or Below	Fair
Lower Secondary	Good
Upper Secondary (Vocational)	Good
Upper Secondary (Academic)	Good
Bachelor	Good
Master	Good
Total	**Good**

Education Level (ISCED)	Happiness
Primary or Below	7.14
Lower Secondary	7.12
Upper Secondary (Vocational)	7.21
Upper Secondary (Academic)	7.11
Bachelor	7.41
Master	7.59
Total	**7.28**

Education Level (ISCED)	Economy Satisfaction
Primary or Below	4.40
Lower Secondary	4.60
Upper Secondary (Vocational)	5.02
Upper Secondary (Academic)	4.66
Bachelor	5.07
Master	5.67
Total	**4.98**

Health by Education Level:

- Shows most common health status for each education level

Happiness by Education Level:

- Shows average happiness (0-10) for each education level

Economy Satisfaction by Education Level:

- Shows satisfaction with national economy (0-10) by education

How to Read:

- Scan down columns to see values for each education level
- Compare values between levels to see if differences exist
- "Total" row shows overall average

---

6. Glossary of Terms

Education Terms

Tertiary Education: Bachelor's and Master's degrees (highest education levels analyzed)

ISCED (International Standard Classification of Education): Framework enabling comparison of education systems across countries with different structures

Education Levels (7 categories):

1. Primary or Below
2. Lower Secondary
3. Upper Secondary General
4. Upper Secondary Vocational

5. Post-Secondary Non-Tertiary
6. Bachelor
7. Master

Statistical Terms

Parity Index: Ratio measuring equality between groups; 1.0 = perfect equality, <1.0 = male advantage, >1.0 = female advantage

Income Decile: Position on 1-10 scale where 1=lowest 10% of incomes, 10=highest 10%; measured relative to national distribution

Gap Measure: Difference in average outcomes between highest education (Bachelor/Master) and lowest education (Primary/Lower Secondary)

Mean/Average: Sum of all values divided by count; used for continuous variables like income, satisfaction, happiness

Mode: Most common category; used for categorical variables like health status ("Good" is most common)

Cross-sectional Data: Data collected at single point in time; shows associations but cannot prove causation

Longitudinal Data: Data tracking same individuals over time; enables causal inference

Quality of Life Terms

Life Satisfaction: Cognitive evaluation of life overall (0-10 scale); measured by question "How satisfied are you with your life as a whole?"

Happiness: Emotional/affective wellbeing (0-10 scale); measured by question "How happy did you feel yesterday?"

Subjective Wellbeing: Self-reported assessments of life quality including satisfaction, happiness, health perceptions

Income Feeling: Subjective assessment of household income adequacy (4 categories: Living Comfortably, Coping, Difficult, Very Difficult)

Country and Policy Terms

SDG 4.5.1: UN Sustainable Development Goal indicator measuring parity indices for access to education across gender, wealth, disability, location, ethnicity dimensions

Welfare State: Government system providing social safety nets (healthcare, unemployment benefits, pensions); Nordic countries (Sweden) have strongest welfare states

European Social Survey (ESS): Cross-national survey measuring attitudes, beliefs, and behaviors across 30+ European countries every 2 years

---

7. Data Sources and Updates

Primary Data Source

European Social Survey (ESS)

- Round 9 (2018): https://ess.sikt.no/en/study/172ac431-2a06-41df-9dab-c1fd8f3877e7
- Round 10 (2020): https://ess.sikt.no/en/study/c00ba7e6-1077-4da8-93f0-a0d8b0c0fb67
- Round 11 (2023): https://ess.sikt.no/en/study/a708eba4-7e9f-4e8e-9a52-adaef1d649e3

Data Collection Methodology

- Face-to-face interviews with random probability samples
- Nationally representative samples aged 18+
- Strict quality controls and standardized questionnaires
- Response rates typically 50-70%

Sample Sizes (Approximate)

- Germany: ~8,000 respondents (across 3 rounds)
- Sweden: ~6,500 respondents
- Serbia: ~6,000 respondents
- Spain: ~6,500 respondents
- **Total:** ~32,000 respondents

---

8. Limitations and Interpretation Guidelines

What This Dashboard CAN Tell You

✅ Gender parity has been achieved in tertiary education attainment
✅ Education level associates with higher income and life satisfaction
✅ Association strength varies dramatically by country context
✅ Patterns have remained relatively stable from 2018-2023

What This Dashboard CANNOT Tell You

❌ **Causation:** Cannot prove education *causes* higher income or satisfaction (only shows association)
❌ **Individual predictions:** Cannot predict specific person's outcomes based on education
❌ **Wealth-based equity:** Cannot assess whether poor children have equal access to tertiary education
❌ **Within-country regional variation:** Data aggregated at national level
❌ **Non-binary gender:** Data limited to male/female categories

Critical Interpretation Notes

Cross-sectional limitation: The data shows that currently highly-educated people currently have higher incomes, but we cannot determine whether:

- Education led to higher income, OR
- Already-wealthy people obtained more education, OR
- Other factors (family connections, intelligence, motivation) caused both education and income

Self-reported measures: Life satisfaction and happiness are subjective. Different cultures may interpret scales differently, affecting cross-national comparisons.

Country selection: Four countries cannot represent all of Europe. Conclusions may not apply to Central Europe, Baltic states, or other regions not analyzed.

Appropriate Uses

✅ Exploratory analysis identifying patterns for further investigation
✅ Generating hypotheses about education-outcome relationships
✅ Comparing education systems across welfare state models
✅ Monitoring gender parity achievement toward SDG 4.5.1 targets
✅ Contextualizing education policy discussions

Inappropriate Uses

❌ Predicting individual-level outcomes
❌ Making causal claims about education's effects
❌ Proving policy interventions work without additional evidence
❌ Generalizing to countries not analyzed
❌ Replacing rigorous policy evaluation studies

- Additional visualizations needed
- Countries to add
- Clarity of explanations
- Technical usability issues

Quick Start Checklist

For new users, follow these steps:

1. ☐ Open dashboard link in web browser
2. ☐ Start on "Gender Inequality" page using navigation bar
3. ☐ Read KPI cards to understand sample characteristics
4. ☐ Observe donut chart showing 50/50 gender split
5. ☐ Try selecting one country using tile slicer at top
6. ☐ Watch how visualizations update with selection
7. ☐ Move to "Income Gap" page via navigation bar
8. ☐ Study main line chart showing education-income gradients
9. ☐ Compare Serbia's steep line vs. Sweden's flat line
10. ☐ Move to "Well-Being" page
11. ☐ Compare life satisfaction patterns to income patterns
12. ☐ Return to any page for deeper exploration
